

一. 计算第二型曲面积分

(1) $\iint_S x^2 y^2 z \, dxdy$, $\Sigma: x^2 + y^2 + z^2 = R^2$ 在平面上的投影

$z = \sqrt{R^2 - x^2 - y^2}$ (上表面)


$\iint_S x^2 y^2 z \, dxdy = \iint_{D_{xy}} x^2 y^2 \sqrt{R^2 - x^2 - y^2} \, dxdy$

$= \iint_{D_{xy}} x^2 y^2 \sqrt{R^2 - x^2 - y^2} \, dxdy$

$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad r \in [0, R], \theta \in [0, 2\pi]$

$\iint_S x^2 y^2 z \, dxdy = \iint_{D_{xy}} r^4 \sin^2 \theta \cos^2 \theta \sqrt{R^2 - r^2} \, dxdy$

$= \int_0^{2\pi} \cos^2 \theta \sin^2 \theta \, d\theta \int_0^R r^4 \sqrt{R^2 - r^2} \, dr$

$= \frac{1}{2} \times \frac{1}{4} \int_0^{2\pi} \sin^2 2\theta \, d\theta \int_0^R r^4 \sqrt{R^2 - r^2} \, dr$

$\int_0^{2\pi} \sin^2 2\theta \, d\theta = \pi$

$= \int_0^R r^4 \sqrt{R^2 - r^2} \, dr$

$= \int_0^R (R^2 - r^2)^2 \sqrt{R^2 - r^2} \, dr$

$= R^4 \int_0^R \sqrt{R^2 - r^2} \, dr + \int_0^R r^2 \sqrt{R^2 - r^2} \, dr - 2R^2 \int_0^R r^4 \sqrt{R^2 - r^2} \, dr$

$= R^4 \left[\frac{2}{3} r^{\frac{3}{2}} \right]_0^R + \frac{2}{5} r^{\frac{5}{2}} \Big|_0^R - 2R^2 \left[\frac{2}{5} r^{\frac{5}{2}} \right]_0^R$

$= R^4 \left(\frac{2}{3} R^{\frac{3}{2}} \right) + \frac{2}{5} R^{\frac{7}{2}} - 2R^2 \left(\frac{2}{5} R^{\frac{5}{2}} \right)$

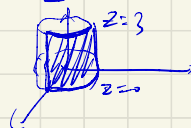
$= R^{\frac{7}{2}} \left(\frac{2}{3} + \frac{2}{5} - \frac{4}{5} \right) = R^{\frac{7}{2}} \frac{2+3-6}{15} = \frac{16}{15} R^{\frac{7}{2}}$

$\therefore \iint_S x^2 y^2 z \, dxdy = \frac{16}{15} R^{\frac{7}{2}}$

$= \frac{2}{15} R^{\frac{7}{2}} \int_0^{2\pi} \sin^2 2\theta \, d\theta$

$= \frac{2}{15} R^{\frac{7}{2}} \left[\theta - \frac{1}{2} \frac{1}{4} \int_0^{2\pi} \cos 4\theta \, d\theta \right] = \frac{2}{15} \pi R^{\frac{7}{2}}$

(2) $\iint_S x \, dxdy + y \, dydz + z \, dzdx$, $\Sigma: x^2 + y^2 = 1$ 在平面上的投影



$\iint_S x \, dxdy + y \, dydz + z \, dzdx = \iint_{D_{xy}} x \, dxdy + \iint_{D_{yz}} y \, dydz + \iint_{D_{xz}} z \, dzdx$

$(\cos \alpha, \cos \beta, \cos \gamma) = \frac{1}{\sqrt{x^2 + y^2}} (x, y, 0)$

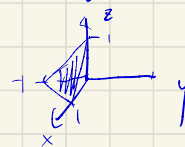
$= \frac{(x, y, 0)}{\sqrt{x^2 + y^2}}$

$\iint_S x \, dxdy + y \, dydz + z \, dzdx = \iint_{D_{xy}} x \frac{x}{\sqrt{x^2 + y^2}} + y \frac{y}{\sqrt{x^2 + y^2}} + 0 \, dxdy$

$= \iint_{D_{xy}} \sqrt{x^2 + y^2} \, dxdy = \iint_{D_{xy}} 1 \, dxdy = \pi \times 1^2 = \pi$

(3) $\iint_S [f(x, y, z) + x] \, dydz + [2f(x, y, z) + y] \, dzdx + [f(x, y, z) + z] \, dxdy$

$\Sigma: x + y + z = 1$ 在第一卦限上的投影



$\Sigma: z = 1 - x - y$

$\frac{\partial z}{\partial x} = -1, \frac{\partial z}{\partial y} = -1$

$\iint_S [f(x, y, z) + x] \, dydz + [2f(x, y, z) + y] \, dzdx + [f(x, y, z) + z] \, dxdy$

$= \iint_{D_{xy}} [f + x - 2f - y + f + 1 - x + y] \, dxdy$

$= \iint_{D_{xy}} 1 \, dxdy = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$

(4) $\oint xz \, dy + xy \, dz + yz \, dx$

$\Sigma: x=0, y=0, z=0, x+y+z=1$ 其中 $z=1-x-y$
 有四个面的面积 $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$



y 在 xy 面上的面积为 $\frac{1}{2}$

$\oint xz \, dy + xy \, dz + yz \, dx$

$\oint_{\Sigma} (xyz \cos \alpha + yz \cos \beta + xz \cos \gamma) \, ds$

$= \frac{1}{\sqrt{3}} \iint_{\Sigma} (xy + yz + xz) \, ds$

$= \frac{1}{\sqrt{3}} \iint_{\Sigma} [xy + (xy) \cdot (1-x-y)] \, dx \, dy = \frac{1}{8}$

三. 求面积

(1) $\begin{cases} x = a \cos^3 t \\ y = a \sin^3 t \end{cases} \quad t \in [0, 2\pi]$

$A = \frac{1}{2} \oint_L x \, dy - y \, dx = \frac{1}{2} \int_0^{2\pi} [a \cos^3 t \cdot 3a \cos^2 t \sin t + a \sin^3 t \cdot 3a \cos^2 t \sin t] \, dt$
 $= \frac{3}{2} a^2 \int_0^{2\pi} [\cos^5 t \sin t + \sin^5 t \cos t] \, dt = \frac{3}{2} a^2 \int_0^{2\pi} \sin t \cos^4 t \, dt = \frac{3}{8} \pi a^2$

(2) $x^2 + y^2 = 2by, b > 0$

设 $\begin{cases} x = b \cos t \\ y = b + b \sin t \end{cases} \quad t: 0 \rightarrow 2\pi$

$A = \frac{1}{2} \oint_L (x \, dy - y \, dx) = \frac{1}{2} \int_0^{2\pi} [b \cos t \cdot b \cos t - (b + b \sin t) \cdot b \cos t] \, dt = \pi b^2$

(3) $(x^2 + y^2)^2 = a^2(x^2 - y^2)$ 求

$\begin{cases} x = a \sqrt{\cos 2\theta} \cos \theta \\ y = a \sqrt{\cos 2\theta} \sin \theta \end{cases}$

$\theta: -\frac{\pi}{2} \rightarrow \frac{\pi}{2}, \frac{\pi}{2} \rightarrow \frac{3\pi}{2}$

$A = \frac{1}{2} \oint_L x \, dy - y \, dx = \frac{1}{2} \left(\int_{-\pi/2}^{\pi/2} a \sqrt{\cos 2\theta} \cos \theta \cdot a \sqrt{\cos 2\theta} \sin \theta \, d\theta - \int_{\pi/2}^{3\pi/2} a \sqrt{\cos 2\theta} \sin \theta \cdot a \sqrt{\cos 2\theta} \cos \theta \, d\theta \right) = a^2$

四. 格林公式

(1) $\oint_L (a_1x + b_1y + c_1) \, dx + (a_2x + b_2y + c_2) \, dy$

$L: (x-1)^2 + (y-1)^2 = 9$ 逆时针方向

$P = a_1x + b_1y + c_1, Q = a_2x + b_2y + c_2$

$\frac{\partial Q}{\partial x} = a_2, \frac{\partial P}{\partial y} = b_1$

由格林公式, $\oint_L P \, dx + Q \, dy = \iint_D (a_2 - b_1) \, dx \, dy = (a_2 - b_1) \pi \cdot 9$

(2) $\oint_L (x^2y \cos x + 2xy \sin x - y^2e^x) \, dx + (x^2 \sin x - 2ye^x) \, dy$

$L: x^{\frac{1}{2}} + y^{\frac{2}{3}} = a^{\frac{2}{3}} (a > 0)$

$P = x^{\frac{1}{2}}y \cos x + 2xy \sin x - y^2e^x, Q = x^2 \sin x - 2ye^x$

$\frac{\partial Q}{\partial x} = 2x \sin x + x^2 \cos x - 2ye^x$

$\frac{\partial P}{\partial y} = x^{\frac{1}{2}} \cos x + 2x \sin x - 2ye^x$

$\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$

$\therefore \oint_L P \, dx + Q \, dy = 0$

(3) $\int_L y \, dx + (\sqrt{\sin y} - x) \, dy$

$L: A(-1,0), B(2,1), C(1,2)$ 逆时针方向

$P = y, Q = \sqrt{\sin y} - x$

$\frac{\partial Q}{\partial x} = -1, \frac{\partial P}{\partial y} = \frac{1}{2\sqrt{\sin y}}$

$\therefore \oint_L P \, dx + Q \, dy = \iint_D \left(\frac{1}{2\sqrt{\sin y}} - 1 \right) \, dx \, dy$

$= \iint_D dy \, dx = 2S$

$= 2 \times \frac{1}{2} \times 2 \times 1 = 2$



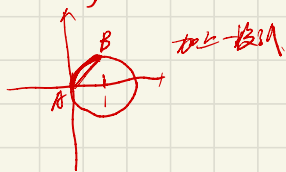
$$(4) \int_L (x^2 - y) dx - (x + \sin^2 y) dy$$

$$L: y = \sqrt{2x - x^2} \text{ 从 } (0,0) \text{ 到 } (1,1) \text{ 的弧.}$$

$$P = x^2 - y, \quad Q = -x - \sin^2 y \quad x^2 + y^2 - 2x = 0$$

$$\frac{\partial Q}{\partial x} = -1$$

$$\frac{\partial P}{\partial y} = -1$$



$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0$$

$$\int_{AB+BA} P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = 0.$$

$$\oint_{AB} P dx + Q dy$$

$$= \int_{AB} (x^2 - y) dx - (x + \sin^2 y) dy$$

$$= \int_0^1 (-1 + \sin^2 y_1) dy + \int_0^1 x^2 dx$$

$$= \frac{1}{4} \sin 2 - \frac{7}{6}.$$

$$\text{五. } \oint_L \frac{y dx - x dy}{2(x^2 + y^2)}, \quad L: (x-1)^2 + y^2 = 2 \text{ 逆时针方向}$$

$$P = \frac{y}{2(x^2 + y^2)}, \quad Q = \frac{-x}{2(x^2 + y^2)}$$



$$\text{当 } x^2 + y^2 \neq 0 \text{ 时, } \frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} = \frac{x^2 - y^2}{2(x^2 + y^2)^2} \Rightarrow \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0.$$

取 $R(0,0)$ 到 $R=100$ 的圆环 E

$$\oint_{L+E} P dx + Q dy = \iint_E \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = 0$$

$$\oint_{L+E} P dx + Q dy \text{ 取 } \begin{cases} x = E \cos \theta \\ y = E \sin \theta \end{cases} \quad \theta \in [0, 2\pi]$$

$$R_E = \oint_L P dx + Q dy$$

$$= \oint \frac{y}{2(x^2 + y^2)} dx + \frac{-x}{2(x^2 + y^2)} dy$$

$$= \oint_0^{2\pi} \frac{E \sin \theta}{2E^2} d(E \cos \theta) + \frac{-E \cos \theta}{2E^2} d(E \sin \theta)$$

$$= \int_0^{2\pi} \frac{1}{2} d\theta = \frac{1}{2} (2\pi - 0) = \pi.$$

$$\therefore R_E \text{ 为 } -\pi.$$